Low Demand Is Not Low Carbon

What 8,760 Hours of German Grid Data Say About When Flexible Load Should Run

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1. A Claim Worth Testing

A few weeks ago I posted about Germany's 2024 grid-congestion bill: €2.9 billion spent managing a system that curtailed 9.4 TWh of renewable generation it had already produced. In my first write-up from that rabbit hole I made a claim in passing, that demand is a poor stand-in for carbon intensity. A passing claim is not evidence, and the carbon-accounting community leans on it constantly, in the form of "shift flexible load to off-peak hours". With some free time on my hands I went back to the hourly data to see whether it holds up.

It does not, at least not in Germany in 2025. What drives the hourly carbon signal is how much wind and solar is on the system, and demand has almost no say in it. That has consequences for how flexible load (EV fleets, heat pumps, batch processes, data centres) should be scheduled and for how it should be reported under Scope 2.

2. Data and Method

The analysis uses 8,760 hourly observations for 2025 from SMARD, the Bundesnetzagentur market data platform: grid load, residual load and generation by technology. Hourly carbon intensity is the sum of each fuel's generation times its emission factor, divided by total generation in that hour. The factors are lignite 1,020, hard coal 820, other conventional 700, fossil gas 490 and biomass 200 gCO_2/kWh; wind, solar, hydro and nuclear carry zero operational emissions. The result is a generation-based, operational estimate that ignores imports and life-cycle emissions. Over the year it averages 331 gCO_2/kWh and ranges from 76 to 665.

Residual load is grid load minus wind and solar output. I call an hour "low-carbon" if it falls in the cleanest 10% of the year (at or below 131 gCO_2/kWh), which gives 876 hours. Correlations are Pearson's r, calculated on the 8,760 hourly values and, where stated, on the 288 month-by-hour average cells.

3. Wind and Solar Output Tracks Carbon Intensity Almost One-for-One

Across the 288 month-by-hour averages, wind plus solar output and carbon intensity have a correlation of r = −0.91. On raw hourly data it is the same, −0.91. A simple fit puts the slope at roughly −9 gCO_2/kWh for every additional gigawatt of wind and solar on the system. Table 1 shows it by quintile: in the fifth of hours with the least wind and solar the grid averages 521 gCO_2/kWh; in the fifth with the most it averages 147, a factor of 3.5.

Table 1. Carbon intensity and grid load by quintile of hourly wind + solar output, 2025.

Wind + solar quintile	Average wind + solar (GW)	Average grid load (GW)	Average carbon intensity (gCO_2/kWh)
1 (lowest)	5.9	50.3	521
2	13.4	51.0	410
3	21.9	53.8	330
4	31.4	55.4	248
5 (highest)	45.3	55.3	147

Look at the grid-load column in the same table. It rises by about 5 GW from the first quintile to the fifth, while carbon intensity falls by nearly three-quarters. Demand is moving a little in the “wrong” direction and the carbon signal ignores it.

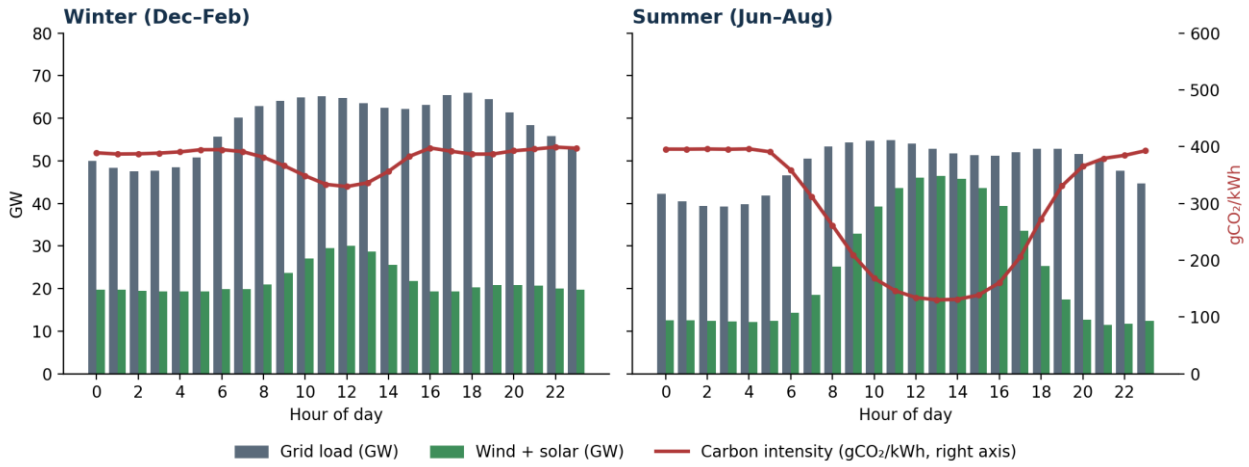


Figure 1. Average grid load, wind + solar output and carbon intensity by hour of day, winter (Dec–Feb) and summer (Jun–Aug), 2025.

Figure 1 shows the mechanism. In summer, solar lifts wind-plus-solar output from about 12 GW overnight to over 45 GW at midday, and carbon intensity drops from roughly 395 to 130 gCO_2/kWh with it. In winter there is little solar to lift anything: output barely exceeds 30 GW at noon and the carbon curve is shallow, dipping only to about 330. Noon is about two and a half times cleaner in summer than in winter.

4. Demand Tells You Almost Nothing About Carbon

On the 8,760 hourly values, grid load and carbon intensity have a correlation of r = 0.03, which is zero for practical purposes. On the month-by-hour averages it is −0.09. Figure 2 puts the two relationships side by side; the left panel is a tight downward band, the right panel is a cloud.

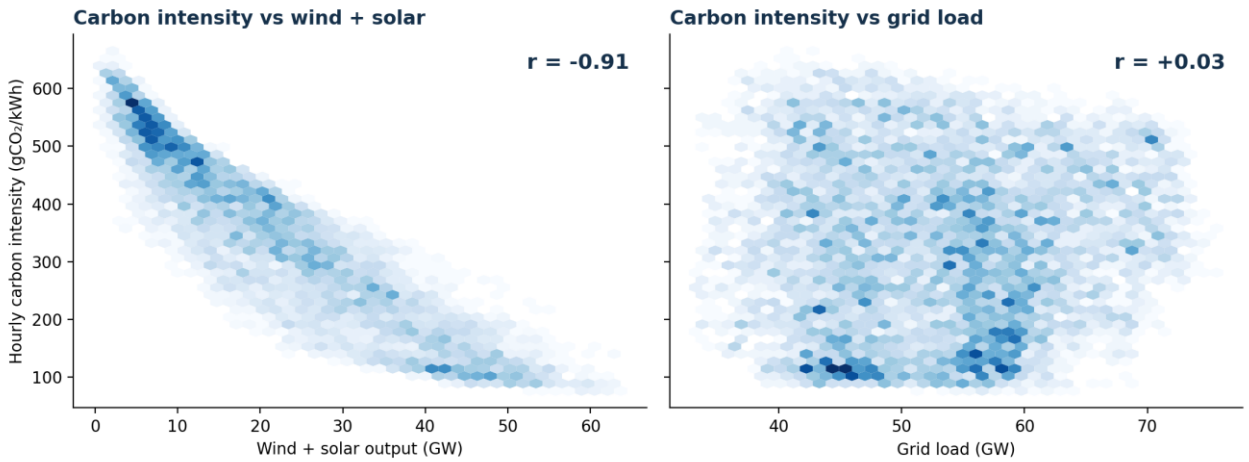


Figure 2. Hourly carbon intensity against wind + solar output (left) and against grid load (right), 2025. Colour shows the number of hours in each cell.

The overnight trough is the clearest illustration. Between 02:00 and 05:00 grid load is at its lowest of the day, about 43 GW. Carbon intensity in those same hours averages 394 gCO_2/kWh, the highest of the day. Between 11:00 and 14:00 load is near its daily maximum, about 58 GW, and carbon intensity averages 226. The cheapest hours to run flexible load are, in carbon terms, among the worst.

The reason is not mysterious once the first finding is in view. Carbon intensity follows what is generating, and demand affects it only indirectly, by changing how much of the fossil fleet is needed. Overnight there is no solar and wind is whatever the weather provides, so the fossil share stays high whatever the load happens to be.

5. What Different Scheduling Rules Actually Deliver

To put numbers on the practical difference, I took an idealised flexible load that can place one-tenth of the year's hours wherever it likes, and asked what carbon intensity it would experience under four rules. The load is assumed to be perfectly flexible and to know the future, so these are upper bounds on what any real scheme could achieve. What matters is how the rules rank.

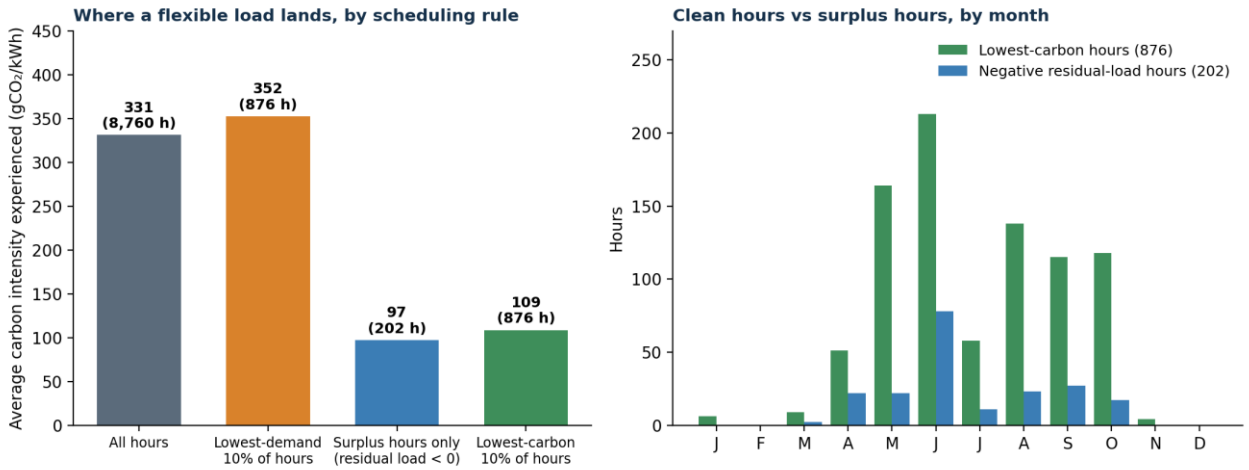


Figure 3. Average carbon intensity experienced under four scheduling rules (left); lowest-carbon hours and negative residual-load hours by month (right), 2025.

Table 2. Average carbon intensity experienced by a flexible load under each scheduling rule.

Scheduling rule	Hours used	Average gCO_2/kWh	Versus all-hours average
No scheduling (all hours)	8,760	331	—
Run in the lowest-demand 10% of hours	876	352	+6%
Run only in surplus hours (residual load < 0)	202	97	−71%
Run in the lowest-carbon 10% of hours	876	109	−67%

The "run when demand is low" rule lands at 352 gCO_2/kWh, about 6% worse than not scheduling at all. Targeting the carbon signal directly cuts the load's exposure by two-thirds. The lowest-demand hours are mostly summer nights, which is exactly where the demand proxy fails.

A second point is easy to miss. The surplus rule does very well per hour, 97 gCO_2/kWh, but it only has 202 hours to work with, 2.3% of the year. There are 876 low-carbon hours, 4.3 times as many, and 674 of them (77%) have positive residual load: wind and solar are plentiful and cheap, and nothing is being wasted, but the grid is still very clean. Those 674 hours average 112 gCO_2/kWh. A flexibility scheme triggered only by surplus would leave three-quarters of the clean hours unused. The right-hand panel of Figure 3 shows how much wider the clean window is than the surplus window in every month from March to October.

6. What This Means for Carbon Accounting and Decarbonisation Analytics

- **Use the carbon signal, not a proxy.** Off-peak tariffs, low-demand windows and "night-time charging" are reasonable for cost and network reasons, but they should not be reported as carbon measures. In this dataset the demand proxy performs slightly worse than random timing.
- **Trigger flexibility on clean hours, not only surplus hours.** Surplus-based schemes are easy to describe and will miss most of the opportunity. A threshold on carbon intensity, or on wind-plus-solar output as a cheap stand-in for it, captures far more.
- **Expect a seasonal split.** From December to February there are essentially no low-carbon hours (six in January, none in the other two months), so winter flexibility is mostly about avoiding the worst hours rather than finding clean ones. Summer flexibility is about concentrating load at midday. A single all-year rule will be wrong for one of the two.
- **This is the Scope 2 argument in operational form.** A location-based annual factor of 331 gCO_2/kWh gives the same footprint to a load that sits in the 109 band and one that sits in the 352 band. Hourly or 24/7 matching approaches (EnergyTag, hourly CFE reporting) are designed to reward that difference. The data supports that design choice.

7. Limitations

The estimate is generation-based and operational; imports, exports and upstream or embodied emissions are not included, and a full LCA treatment would add them. Emission factors are representative values, not plant-specific. The 10% threshold is defined on this year's own distribution, and the scheduling comparison assumes perfect foresight and unconstrained flexibility, so it shows a ranking and not a forecast of savings. The analysis is for one country and one year; I would expect the pattern to hold in systems with large solar and wind shares but have not tested it. Price data is not included, so the carbon-optimal and cost-optimal schedules may differ.

8. Conclusion

In 2025 the German grid's hourly carbon intensity was explained far better by wind and solar output (r = −0.91) than by demand (r = 0.03). A rule of thumb that treats low demand as clean power would have scheduled flexible load into hours that were, on average, dirtier than the year as a whole, and a rule built only around surplus would have missed three-quarters of the clean hours. Both are fixed by using the carbon signal itself. This note sits alongside the earlier analysis of the 8.7× annual range in hourly carbon intensity and the congestion figures from the original post. I would be glad to compare notes with anyone working on hourly carbon accounting, flexibility or grid analytics.

Methodology, hourly dataset and Excel workbook available on request.

Data: SMARD (Bundesnetzagentur) hourly consumption and generation, 2025; Bundesnetzagentur congestion-management reporting.